Narges Zarnaghinaghsh

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Summary

- Strong work ethic and initiative leveraged to develop and evaluate biological models, biomolecule design for RNA sequences, and large language models, showcasing strong analytical skills and a broad understanding of multimedia processing techniques, as evidenced by 13 documents.
- Self-motivated with an ability to work autonomously pioneering innovative biomolecule design, biological models models, generative models, and tumor detection techniques using compressive sensing, as evidenced by 9 optimized methodologies.
- Strong communication skills with proficiency in information synthesis, interpretation and delivery, resulting in 5 presentations.
- Cross-functional collaborator with experience developing different generative models such as RNA grammar variational autoencoder (RGVAE) for RNA, modified 5'UTR, and mRNA design and optimization, including optimizing the model with desired Ribosome Load (RL), MFE, GC-content, RNA length, structure, mandatory and forbidden motifs, as evidenced by 9 publications including 1 in Bioinformatics Advances Journal.
- Experience in managing project details and tasks using proficiency in deep learning models implementation, ML Framework, complex algorithm design, signal processing and biomolecule design as evidenced by 29 projects.
- Proficiency in applying deep learning techniques to implement various scientific models and biomolecule design showcasing versatility in adapting to different machine learning architectures using ResNet, Transformers, Image Classification, and Autoencoders, as demonstrated by working with 9 teams.

Work Experience

Computer Networks, Microwave Imaging and Modeling Skills
Gained as Research Assistant at Texas A&M University, Lecturer at Texas A&M University and Teaching Assistant at Amirkabir University

- Exceptional system integration abilities, research experience, and collaboration skills showcased by combining SDN with NDN as evidenced by 8 collaborations.
- Teaching and implementing metropolitan area networks to about 300 undergraduate students.
- Teaching microwave Lab to about 15 undergraduate students.

Data Scientist, Computer Networks and AI trainer Skills
Gained as Data Scientist at Spectral MD Company, AI trainer at Mercor and AI Trainer and Evaluator at Handshake AI Move

- High-level technical, statistical, analytical, and coding skills with proficiency utilizing computer networks, C/C++, C#, LATEX, LABVIEW, MATLAB, VHDL, Python, JAVA, R statistics, PyTorch, Tensorflow, Keras, Pandas, Amazon AWS, and SQL for advanced data storytelling, as demonstrated by implementing 16 software tools.
- Experience in training frontier AI models by engineering prompts for stumping these models, and developing codes as a professional electrical engineer and AI experts.
- Experience in data cleaning, high-performance computing, reinforcement learning, applying deep learning for medical image classification, scientific models, and evaluating the ML model as a professional AI trainer for large language models.

Education

M.Sc. in Electrical Engineering
Amirkabir University of Technology

B.Sc. in Electrical Engineering
Amirkabir University of Technology

PhD candidate in Electrical Engineering
Texas A&M University

Skills List

Machine Learning Scientist I/II, Biomolecule Design research experience
Pytorch
Coding Skills
AI trainer
Bioinformatics
Deep Learning
Generative Models
Java, C++, C, Python MATLab
Machine learning
Presentation Skills
Reinforcement Learning
Stakeholder Communication
Variational Autoencoders

Affiliations & Hobbies

- Ranked 1st in a national Quran memorization competition in the US.
- Handshake badge for the successful completion of Model Validation 2 - Expert.
- Publications: https://scholar.google.com/citations?user=o_eDff8AAAAJ&hl=en
- Best Paper Award at 1st National Conference on Development of Civil Engineering, Architecture, Electricity and Mechanical in Iran